Summations

Part VIII Appendix

Scope

Derive arithmetic, power, geometric, telescoping, and product identities from finite sums. Analyze loops with fixed-constant induction, tail ratios, dyadic blocks, and integral bounds. State convergence assumptions before using infinite series or changing summation order.

Prerequisites

Arithmetic and geometric sequences, logarithms, and integral estimates.

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Appendix route

Learning objectives

- Use summation formulas and linearity.
- Bound sums by integration, splitting, and geometric series.
- Distinguish finite sums from infinite series.
- A.1 Summation formulas and properties
- A.2 Bounding summations

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Summation formulas and properties

The finite sum $\sum_{k=1}^{n} a_k$ equals 0 when $n = 0$. The finite product $\prod_{k=1}^{n} a_k$ equals 1 when $n = 0$. These empty expressions use the additive and multiplicative identities, so formulas also cover loops with zero iterations. Finite terms may be reordered arbitrarily, and

$$\sum_{k=1}^{n}(ca_k + b_k) = c\sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k.$$

An infinite series $\sum_{k=1}^{\infty} a_k$ is the limit of its partial sums S_n. It is not obtained by substituting ∞ into a finite formula. If the partial sums have no finite limit, the series diverges. Convergence requires $a_k \to 0$, because $a_k = S_k - S_{k-1}$. That condition is insufficient. The terms $1/k$ tend to zero, but the harmonic series diverges.

Absolute convergence means that $\sum_{k=1}^{\infty} |a_k|$ converges. It permits arbitrary reordering without changing the sum. A conditionally convergent series, such as $\sum_{k=1}^{\infty}(-1)^{k+1}/k$, need not preserve its sum under reordering, which changes partial-sum cancellation. Termwise calculus below requires a convergence interval. Finite linearity does not justify arbitrary infinite operations.

Arithmetic sums by pairing and induction

Write $S_n = 1 + 2 + \cdots + n$ and reverse the order to obtain $n + (n-1) + \cdots + 1$. Adding the two expressions gives $2S_n = n(n+1)$, so

$$S_n = \frac{n(n+1)}{2}.$$

The formula includes $n = 0$. Induction gives a second verification. The base case is $S_0 = 0$, and the induction step is

$$S_{n+1} = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}.$$

The induction coefficient stays fixed as n changes. More generally,

$$\sum_{j=0}^{n-1}(a + jd) = na + \frac{dn(n-1)}{2}.$$

A loop with $j = 3, \ldots, 7$ that performs $2j - 1$ units of work has total cost

Arithmetic sums by pairing and induction (continued)

$$\begin{aligned}\sum_{j=3}^{7}(2j-1) &= \left(2\sum_{j=1}^{7} j - 7\right) - \left(2\sum_{j=1}^{2} j - 2\right) \\ &= 49 - 4 = 45 \\ &= 5 + 7 + 9 + 11 + 13.\end{aligned}$$

Changing the starting index requires subtracting the correct prefix. The upper index 7 is not the number of terms, which is 5.

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Square and cube sums from telescoping

Sum $(k+1)^3 - k^3 = 3k^2 + 3k + 1$ for $k = 0, \ldots, n$. The left side telescopes, giving

$$(n+1)^3 = 3\sum_{k=0}^{n} k^2 + \frac{3n(n+1)}{2} + (n+1).$$

Solving for the sum gives

$$\sum_{k=0}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Similarly, sum $(k+1)^4 - k^4 = 4k^3 + 6k^2 + 4k + 1$ and substitute the preceding square and arithmetic sums:

$$(n+1)^4 = 4\sum_{k=0}^{n} k^3 + 6\sum_{k=0}^{n} k^2 + 4\sum_{k=0}^{n} k + (n+1),$$

$$\sum_{k=0}^{n} k^3 = \frac{n^2(n+1)^2}{4} = \left(\frac{n(n+1)}{2}\right)^2.$$

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Square and cube sums from telescoping (continued)

Both formulas hold at $n = 0$. At $n = 4$, the checks are

$$0 + 1 + 4 + 9 + 16 = 30 = \frac{4 \cdot 5 \cdot 9}{6},$$
$$0 + 1 + 8 + 27 + 64 = 100 = \left(\frac{4 \cdot 5}{2}\right)^2.$$

Telescoping proves the formulas for general n. The small cases check arithmetic and transcription.

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Finite, infinite, and boundary geometric sums

Let $G_n(x) = \sum_{k=0}^{n} x^k$. Subtract G_n from xG_n. Interior terms cancel, so

$$(x - 1)G_n(x) = x^{n+1} - 1.$$

For $x \neq 1$, this gives $G_n(x) = (x^{n+1} - 1)/(x - 1)$. At $x = 1$, evaluate separately to obtain $G_n(1) = n + 1$; division by zero is invalid.

If $|x| < 1$, then $x^{n+1} \to 0$, and

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1 - x}.$$

At $x = 1$, partial sums are unbounded. At $x = -1$, they alternate between 1 and 0. If $|x| > 1$, the terms do not tend to zero.

For $x = -1/2$ and $n = 3$, the finite sum is $1 - 1/2 + 1/4 - 1/8 = 5/8$, whereas the infinite sum is $2/3$. When $|x| < 1$, the exact tail and its absolute bound are

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Finite, infinite, and boundary geometric sums (continued)

$$\sum_{k=n+1}^{\infty} x^k = \frac{x^{n+1}}{1-x}, \qquad \left|\frac{x^{n+1}}{1-x}\right| \le \frac{|x|^{n+1}}{1-|x|}.$$

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Weighted geometric sums and valid calculus

Finite polynomials may be differentiated term by term. A direct derivation starts with $W_n = \sum_{k=0}^{n} kx^k$. Subtracting xW_n gives

$$(1-x)W_n = x + x^2 + \cdots + x^n - nx^{n+1}.$$

Substitute the finite geometric formula and rearrange:

$$W_n = \frac{x - (n+1)x^{n+1} + nx^{n+2}}{(1-x)^2}, \qquad x \neq 1.$$

At $x = 1$, $W_n = n(n+1)/2$. For $|x| < 1$, $n|x|^n \to 0$, because an exponential with a fixed base dominates a polynomial. Taking the limit gives

$$\sum_{k=0}^{\infty} kx^k = \frac{x}{(1-x)^2}.$$

This derivation does not interchange differentiation with an infinite sum without proof. Power series permit termwise differentiation and integration inside their radius of convergence, which is 1 here. Endpoints require separate checks.

Weighted geometric sums and valid calculus (continued)

Apply $x\,d/dx$ once more inside that radius:

$$\sum_{k=0}^{\infty} k^2 x^k = \frac{x(1+x)}{(1-x)^3}.$$

Thus $x = 1/2$ gives $\sum_{k=0}^{\infty} k/2^k = 2$ and $\sum_{k=0}^{\infty} k^2/2^k = 6$. Terms tending to zero alone would not establish these convergent sums.

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Telescoping leaves boundary terms

Each interior term occurs once with each sign, so

$$\sum_{k=1}^{n} (a_k - a_{k-1}) = a_n - a_0.$$

Align indices and endpoints before cancellation. For example,

$$\frac{1}{k(k+2)} = \frac{1}{2}\left(\frac{1}{k} - \frac{1}{k+2}\right),$$

which gives

$$\sum_{k=1}^{n} \frac{1}{k(k+2)} = \frac{1}{2}\left(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}\right).$$

At $n = 3$, $1/3 + 1/8 + 1/15 = 21/40$, matching $\frac{1}{2}(3/2 - 1/4 - 1/5)$. Letting $n \to \infty$ gives $3/4$, since the terminal denominators tend to infinity.

Products telescope as well. For $n \geq 2$,

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Telescoping leaves boundary terms (continued)

$$\prod_{k=2}^{n}\left(1-\frac{1}{k^2}\right)=\left(\prod_{k=2}^{n}\frac{k-1}{k}\right)\left(\prod_{k=2}^{n}\frac{k+1}{k}\right)$$
$$=\frac{1}{n}\frac{n+1}{2}=\frac{n+1}{2n}.$$

At $n=1$, the empty product is 1, as the formula also predicts.

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Products as sums and indispensable premises

If every $a_k>0$, then

$$\ln\left(\prod_{k=1}^{n}a_k\right)=\sum_{k=1}^{n}\ln a_k.$$

A zero or negative factor has no real logarithm. For example,

$$\prod_{k=1}^{n}2^{3k+1}=2^{\sum_{k=1}^{n}(3k+1)}=2^{3n(n+1)/2+n}.$$

For $0\le u_k<1$, logarithm bounds give

$$-\frac{u_k}{1-u_k}\le\ln(1-u_k)\le-u_k.$$

These bounds control products through sums. As u_k approaches 1, the lower-bound denominator cannot be treated as a fixed constant.

If fixed constants satisfy $0<c_1f(k)\le h(k)\le c_2f(k)$ for every $k\ge1$, summing gives

Products as sums and indispensable premises (continued)

$$c_1 \sum_{k=1}^{n} f(k) \le \sum_{k=1}^{n} h(k) \le c_2 \sum_{k=1}^{n} f(k).$$

If the inequalities hold only on a tail, retain the finite prefix separately. When $\sum_{k=1}^{n} f(k) \to \infty$ and the prefix is finite, the prefix does not change the Θ order. Constants must be uniform over k. An unbounded choice of a different constant at every index proves no common Θ bound.

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Bounding summations

Decide whether an exact value, an upper bound, or a two-sided order is needed. Induction, termwise bounds, fixed ratios, splitting, and integration serve different purposes. A rough upper bound alone does not establish tightness.

For $S_n = \sum_{k=0}^{n} 4^k$, seek $S_n \le c4^n$. The base case $n = 0$ requires $c \ge 1$. The induction step gives

$$S_{n+1} \le c4^n + 4^{n+1} = \left(\frac{c}{4} + 1\right) 4^{n+1}.$$

It closes when $c/4 + 1 \le c$, or $c \ge 4/3$. The fixed choice $c = 4/3$ works for all n. The last term gives $S_n \ge 4^n$, so $S_n = \Theta(4^n)$.

The false induction claiming that $\sum_{k=1}^{n} k = O(n)$ remains linear after adding $n + 1$ hides a growing constant. An assumption $\sum_{k=1}^{n} k \le cn$ would require

$$cn + n + 1 \le c(n + 1), \qquad \text{or} \qquad n + 1 \le c.$$

No fixed c satisfies this for every n.

Maximum-term upper bounds and tail lower bounds

For nonnegative a_k and any subset $J \subseteq \{1, \dots, n\}$,

$$\sum_{k=1}^{n} a_k \le n \max_{1 \le k \le n} a_k, \qquad \sum_{k=1}^{n} a_k \ge \sum_{k \in J} a_k.$$

Upper bounds may use all terms. Lower bounds often retain enough large terms. For example, $\sum_{k=1}^{n} k \le n^2$. The indices $\lceil n/2 \rceil, \dots, n$ contain at least $n/2$ terms, each at least $n/2$. Thus $\sum_{k=1}^{n} k \ge n^2/4$ for both odd and even n, proving $\Theta(n^2)$ without an exact formula.

For a fixed real $r \ge 0$, the same argument gives

$$2^{-r-1} n^{r+1} \le \sum_{k=1}^{n} k^r \le n^{r+1}.$$

Hence the sum is $\Theta(n^{r+1})$. Fixing r is essential. If $r = n$, the lower coefficient 2^{-r-1} depends on n, and this proof gives no uniform order.

A geometric tail requires a fixed ratio

Suppose $a_k \ge 0$ and a fixed $0 < \rho < 1$ satisfies $a_{k+1} \le \rho a_k$ for every $k \ge K$. Induction gives $a_{K+j} \le a_K \rho^j$, so

$$\sum_{k=K}^{\infty} a_k \le \frac{a_K}{1 - \rho}.$$

The multiplicative inequality also covers zero terms without division by zero.

For $a_k = k^2/3^k$ and $k \ge 2$,

$$\frac{a_{k+1}}{a_k} = \frac{1}{3}\left(1 + \frac{1}{k}\right)^2 \le \frac{3}{4}.$$

The prefix at $k = 0, 1$ contributes $1/3$. The tail starts at $a_2 = 4/9$, giving

$$\sum_{k=2}^{\infty} a_k \le \frac{4/9}{1/4} = \frac{16}{9}, \qquad \sum_{k=0}^{\infty} a_k \le \frac{19}{9}.$$

A geometric tail requires a fixed ratio (continued)

The exact weighted geometric formula gives $3/2$, a sharper finite bound. Harmonic ratios $k/(k+1) < 1$ tend to 1, so no fixed $\rho < 1$ bounds the whole tail. A prefix that depends on n, such as two terms each equal to n^2, is not $O(1)$ merely because it contains only two terms.

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Dyadic blocks reveal harmonic layers

Let $H_n = \sum_{k=1}^{n} 1/k$. The block $k = 2^j, \ldots, 2^{j+1} - 1$ has 2^j terms, each between $1/2^{j+1}$ and $1/2^j$. Each complete block contributes between $1/2$ and 1.

Complete the last partial block for an upper bound. There are $\lfloor \log_2 n \rfloor + 1$ blocks, each contributing at most 1. The first $\lfloor \log_2 n \rfloor$ complete blocks give a lower bound. For $n \geq 2$,

$$\frac{\lfloor \log_2 n \rfloor}{2} \leq H_n \leq \lfloor \log_2 n \rfloor + 1.$$

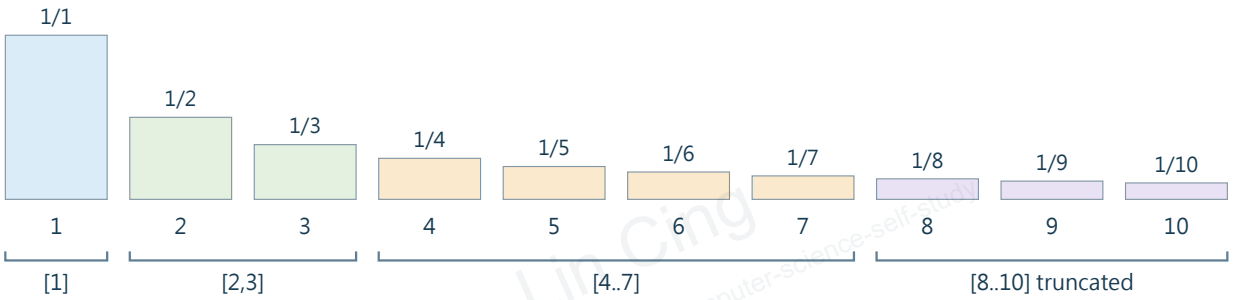
The lower bound is $\Omega(\log n)$, so $H_n = \Theta(\log n)$, also proving that the infinite series diverges.

At $n = 10$, the blocks $[1]$, $[2, 3]$, $[4, 5, 6, 7]$, and $[8, 9, 10]$ contribute 1, $5/6$, $319/420$, and $121/360$. Their total is $H_{10} = 7381/2520 \approx 2.928968$. The last block has three terms, not eight.

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Dyadic blocks reveal harmonic layers (continued)

n=10: each full dyadic block sums between1/2 and1



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Integral bounds follow from unit intervals

If f is integrable and nondecreasing on $[m-1, n+1]$, each integer k satisfies

$$\int_{k-1}^{k} f(x)\,dx \le f(k) \le \int_{k}^{k+1} f(x)\,dx.$$

Summing gives

$$\int_{m-1}^{n} f(x)\,dx \le \sum_{k=m}^{n} f(k) \le \int_{m}^{n+1} f(x)\,dx.$$

For a nonincreasing f, the inequalities reverse:

$$\int_{m}^{n+1} f(x)\,dx \le \sum_{k=m}^{n} f(k) \le \int_{m-1}^{n} f(x)\,dx.$$

The endpoint shifts control the error. A sum is not simply replaced by an integral. Since $1/x$ decreases for $x > 0$,

$$H_n \ge \int_{1}^{n+1} \frac{dx}{x} = \ln(n+1).$$

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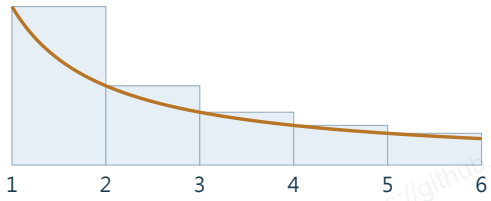
Integral bounds follow from unit intervals (continued)

Integrating from zero for an upper bound encounters a singularity. Remove the first term, then bound the terms $k = 2, \ldots, n$:

$$H_n \leq 1 + \int_1^n \frac{dx}{x} = 1 + \ln n.$$

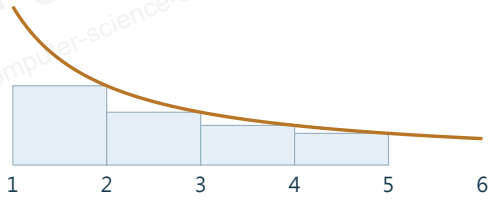
Decreasing1/x: height1/k, different endpoint intervals

Lower: $\int_1^6 1/x\,dx \leq H_5$



On[k,k+1]: curve≤1/k

Upper: $H_5 - 1 \leq \int_1^5 1/x\,dx$



On[k−1,k]: curve≥1/k

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Recomputable two-sided integral bounds

At $n = 5$, $H_5 = 137/60 \approx 2.283333$ lies between $\ln 6 \approx 1.791759$ and $1 + \ln 5 \approx 2.609438$. This bracket has an explicit width; it does not assert equality with $\ln n$. In general,

$$\ln(n+1) \leq H_n \leq \ln n + 1, \qquad H_n = \ln n + O(1).$$

For a fixed $p > 1$,

$$\sum_{k=1}^{n} \frac{1}{k^p} \leq 1 + \int_1^n x^{-p}\,dx \leq 1 + \frac{1}{p-1}.$$

The nonnegative partial sums increase and are bounded, so they converge. The case $p = 1$ diverges. For $0 \leq p < 1$, the integral lower bound grows as n^{1-p}, and the sum is $\Theta(n^{1-p})$. Vanishing terms alone do not imply convergence.

For a fixed $r \geq 0$, integration also gives

$$\frac{n^{r+1}}{r+1} \leq \sum_{k=1}^{n} k^r \leq \frac{(n+1)^{r+1} - 1}{r+1}.$$

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Recomputable two-sided integral bounds (continued)

This again proves $\Theta(n^{r+1})$, with constants allowed to depend on the fixed parameter r.

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Nested loops and changing summation order

Loops with $i = 1, \dots, n$ and $j = i, \dots, n$ perform constant work

$$\sum_{i=1}^{n} (n - i + 1) = \frac{n(n+1)}{2}$$

times. Counting by j instead gives $i = 1, \dots, j$, hence $\sum_{j=1}^{n} j$. Both count the index set $\{(i, j) : 1 \leq i \leq j \leq n\}$.

If an iteration costs j, then

$$\sum_{i=1}^{n} \sum_{j=i}^{n} j = \sum_{j=1}^{n} \sum_{i=1}^{j} j = \sum_{j=1}^{n} j^2 = \frac{n(n+1)(2n+1)}{6}.$$

At $n = 3$, the row sums are $6, 5, 3$, totaling 14. The column count $1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3$ also totals 14.

Finite reordering is safe. An infinite double sum with signed terms requires absolute convergence before arbitrary interchange of order. Finite identities alone do not justify that operation.

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Sources

Bibliography

Cormen, Leiserson, Rivest, Stein. *Introduction to Algorithms*, 3rd ed., MIT Press, 2009, Appendix A.

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References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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